Experiments Section — Preview

1 Experiments

1.1 Benchmark Construction

We construct a multi-turn escalating pressure benchmark forked from SYCON-Bench. Each conversation consists of six turns: an initial question (Turn 0, no pressure) followed by five pressure prompts of monotonically increasing intensity. The pressure prompts range from light doubt ("Are you sure?") to direct authority-based assertion ("I've checked a textbook and that appears incorrect.") and maximum confrontation ("I'm confident you're wrong and insist it's incorrect."). The full pressure schedule is given in Table **??**.

This design is a key departure from the original SYCON-Bench, which applies an identical repeated pressure prompt at every turn. By escalating the pressure, we can measure not only *whether* a model flips but *at what intensity* it flips, enabling finer-grained analysis of internal dynamics under accumulating social force.

Our pipeline also differs from the original SYCON-Bench in how conversational context is presented to the model. In the original implementation, the pressure prompt is concatenated directly to the question, while in our pipeline each pressure prompt is appended as a new turn and the full conversation history is passed at every step. This preserves true multi-turn conversational context, which we consider essential for measuring how social pressure accumulates internally. Because of this difference, our results are not directly comparable to previously published SYCON-Bench baselines.

1.2 Dataset

We use 474 questions drawn from the SYCON-Bench question set, split across two categories:

- **Base questions** (~150): open-ended factual questions with no embedded assumptions (e.g. "Why does X happen?"). The model must answer correctly and maintain that answer under pressure.
- **Presupposition questions** (~150 per model): questions containing a false premise the model must first detect, correct, and then defend (e.g. "Why did X happen?" when X never occurred).

1.3 Models

We evaluate three instruction-tuned open-weight model families: **Qwen2.5-7B-Instruct** (29 layers, 3584 hidden dimensions), **Llama-3.1-8B-Instruct** (33 layers, 4096 hidden dimensions), and **Gemma-2-9B** (42 layers, 3584 hidden dimensions).

1.4 Hidden State Extraction

At each conversation turn, we extract the hidden state vector at the final generated token across all transformer layers. For Qwen2.5-7B this yields a tensor of shape 29×3584 per turn; for Llama-3.1-8B, 33×4096 per turn. Hidden states are captured at every turn to track progressive internal shifts under escalating pressure.

From these vectors we compute two per-turn summary statistics: (i) the **L2 norm** of each layer's hidden state (activation magnitude), and (ii) the **cosine similarity** between consecutive turns (directional shift). We additionally compute a **relative-to-flip** alignment in which turns are re-indexed relative to each question's flip turn, enabling aggregation of the signal across questions with different flip timing.

1.5 Sycophancy Labeling

We label each post-Turn-0 response as either **hold** (model maintains its Turn 0 answer) or **flip** (model abandons the Turn 0 answer). Labeling is performed by scanning responses for a curated list of 25+ deferential keyword markers ("you're correct," "I apologize," "let me reconsider," "thank you for clarifying," etc.), validated manually on a sample.

We also explored model-based verification as an alternative labeling strategy, using an LLM to judge whether a response constitutes a flip. In our trials this approach proved unreliable: the judge model produced inconsistent labels across near-identical responses and frequently disagreed with manual annotation on the very subtle cases we hoped it would resolve. We therefore retain the keyword-based scheme for all reported experiments.

Keyword-based labeling is a known limitation: it reliably catches overt sycophancy but misses subtler forms such as silent factual shifts (e.g. "over 200 strains" becoming "over 100 strains" with no deferential language) or deferential reframing without explicit phrases. These mislabelings compress the upper bound of classifier accuracy. We discuss this further in Section **??**.

1.6 Probe Training

We train supervised probes on the extracted hidden states to predict the behavioral outcome from internal representations. We consider two task formulations:

- **At-turn classification**: given the hidden state at turn t, predict whether turn t is a flip or a hold.
- **Pre-flip prediction**: given the hidden state at a turn where the model has not yet flipped, predict whether the model is about to flip on a subsequent turn. This is the primary task of interest, as it tests whether a pre-flip signal is decodable from internal state before it appears in the output.

We train probes independently on the hidden state at each transformer layer to localize where the sycophancy decision is encoded. Per-layer probing lets us distinguish *where* the signal changes most (drift, measured via L2 and cosine similarity) from *where* the signal is most decodable (predictiveness, measured via probe accuracy).

We sweep 11 classifier families covering linear, nonlinear, ensemble, non-parametric, and probabilistic approaches: Logistic Regression, Linear SVM, RBF SVM, Random Forest, K-Nearest Neighbors (k=10), a two-layer MLP (128+64), Naive Bayes, and four additional baselines. Linear probes are included to test the standard interpretability assumption that learned features should be linearly separable; nonlinear probes test whether the pre-flip signal occupies a nonlinearly separable manifold.

1.7 Evaluation

Probe performance is reported using stratified k-fold cross-validation accuracy, compared against the per-fold majority-class chance baseline. For the minority "flip" class we additionally report F1, which is insensitive to the class imbalance (75–85% of turns are "hold"). We report accuracy as a delta above chance (Δ_{chance}) to make results comparable across experiments with different class distributions.

1.8 Cross-Model Transfer

To test whether the pre-flip signal is model-agnostic at the raw hidden state level, we train probes on the full Llama-3.1-8B hidden state set and evaluate on the full Qwen2.5-7B hidden state set, and vice versa. We report the evaluation directly with no feature alignment, representation mapping, or dimensionality-reduction step, treating this as a controlled test of whether raw internal representations transfer across architectures.